

The two dots ❶ indicate that two tests passed, which is also clear from the last line of output. This is great! We now know that the function still works for names like Janis Joplin, and we can be confident that it will work for names like Wolfgang Amadeus Mozart as well.

TRY IT YOURSELF

11-1. City, Country: Write a function that accepts two parameters: a city name and a country name. The function should return a single string of the form *City, Country*, such as *Santiago, Chile*. Store the function in a module called *city_functions.py*, and save this file in a new folder so pytest won't try to run the tests we've already written.

Create a file called *test_cities.py* that tests the function you just wrote. Write a function called `test_city_country()` to verify that calling your function with values such as 'santiago' and 'chile' results in the correct string. Run the test, and make sure `test_city_country()` passes.

11-2. Population: Modify your function so it requires a third parameter, *population*. It should now return a single string of the form *City, Country - population xxx*, such as *Santiago, Chile - population 5000000*. Run the test again, and make sure `test_city_country()` fails this time.

Modify the function so the *population* parameter is optional. Run the test, and make sure `test_city_country()` passes again.

Write a second test called `test_city_country_population()` that verifies you can call your function with the values 'santiago', 'chile', and 'population =5000000'. Run the tests one more time, and make sure this new test passes.

Testing a Class

In the first part of this chapter, you wrote tests for a single function. Now you'll write tests for a class. You'll use classes in many of your own programs, so it's helpful to be able to prove that your classes work correctly. If you have passing tests for a class you're working on, you can be confident that improvements you make to the class won't accidentally break its current behavior.

A Variety of Assertions

So far, you've seen just one kind of assertion: a claim that a string has a specific value. When writing a test, you can make any claim that can be expressed as a conditional statement. If the condition is `True` as expected, your assumption about how that part of your program behaves will be confirmed; you can be confident that no errors exist. If the condition you